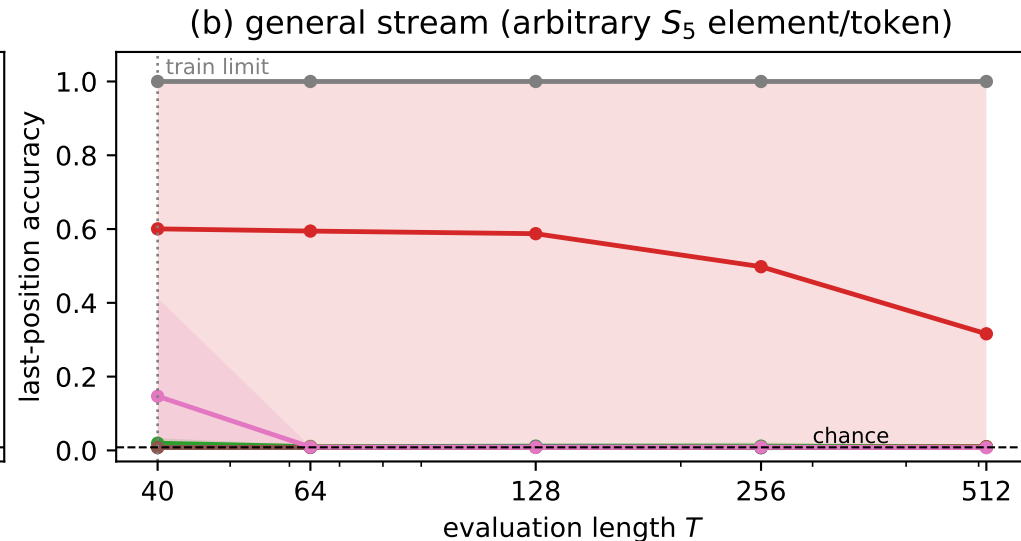
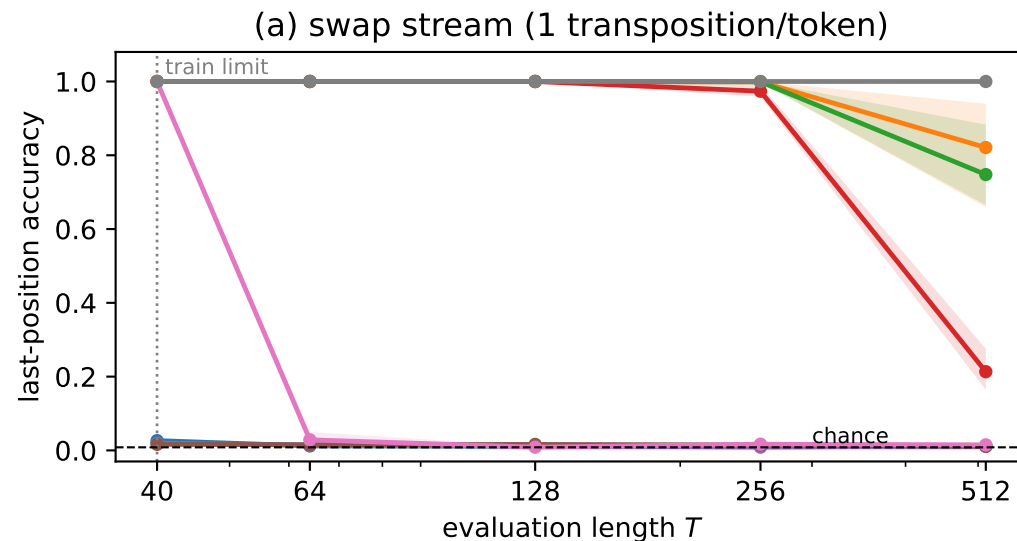


Exp C (P2): state-tracking bifurcation on S_5 — measured



● DeltaNet $\beta \in (0, 1)$
● DeltaNet $\beta \in (0, 2)$
● DeltaProduct $n_h = 2$
● DeltaProduct $n_h = 4$
● diagonal SSM
 ● Transformer 2L
 ● Transformer 8L
 ● LSTM

(c) swap: mean accuracy over seeds

	40	64	128	256	512
DeltaNet $\beta \in (0, 1)$	0.03	0.01	0.01	0.01	0.01
DeltaNet $\beta \in (0, 2)$	1.00	1.00	1.00	1.00	0.82
DeltaProduct $n_h = 2$	1.00	1.00	1.00	1.00	0.75
DeltaProduct $n_h = 4$	1.00	1.00	1.00	0.97	0.21
diagonal SSM	0.02	0.01	0.01	0.01	0.01
Transformer 2L	0.02	0.02	0.02	0.01	0.01
Transformer 8L	1.00	0.03	0.01	0.02	0.01
LSTM	1.00	1.00	1.00	1.00	1.00
evaluation length T	40	64	128	256	512

(d) general: mean accuracy over seeds

	40	64	128	256	512
DeltaNet $\beta \in (0, 1)$	0.01	0.01	0.01	0.01	0.01
DeltaNet $\beta \in (0, 2)$	0.01	0.01	0.01	0.01	0.01
DeltaProduct $n_h = 2$	0.02	0.01	0.01	0.01	0.01
DeltaProduct $n_h = 4$	0.60	0.59	0.59	0.50	0.32
diagonal SSM	0.01	0.01	0.01	0.01	0.01
Transformer 2L	0.01	0.01	0.01	0.01	0.01
Transformer 8L	0.15	0.01	0.01	0.01	0.01
LSTM	1.00	1.00	1.00	1.00	1.00
evaluation length T	40	64	128	256	512